Chapter 13

Vectors I

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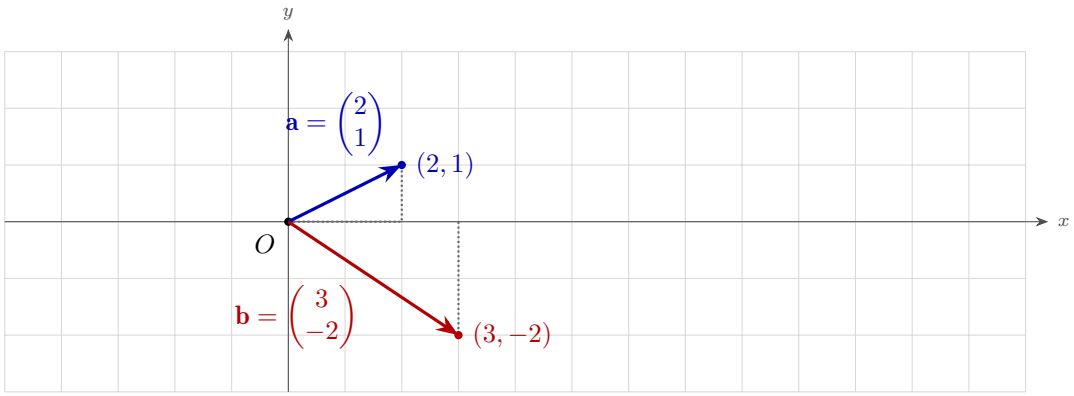
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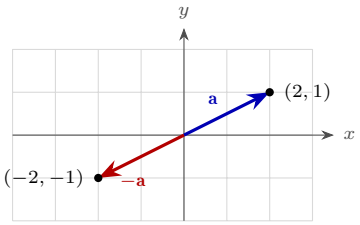
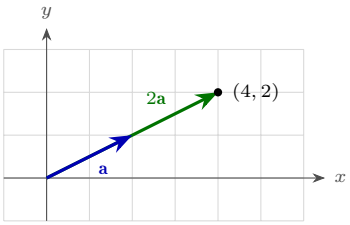
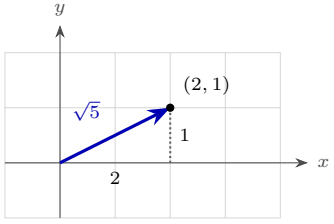
13.1 Vectors in Two Dimensions

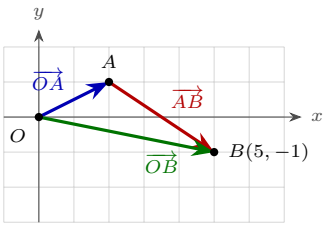
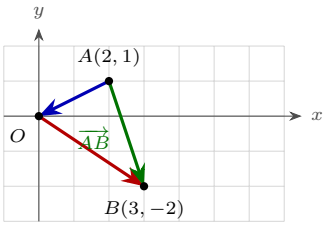
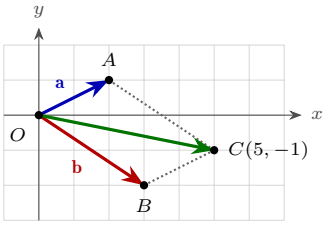
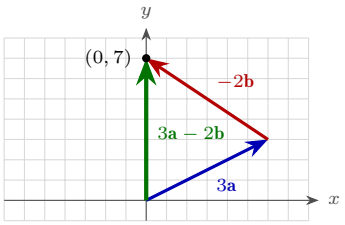
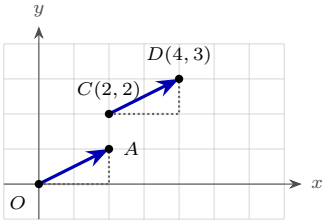
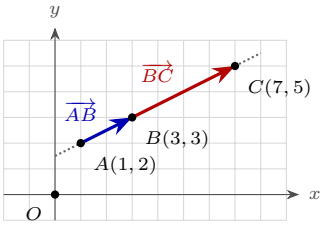
In Secondary 4, we have learned vectors in two dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$$

A vector describes a quantity that has both magnitude and direction. Vectors let us describe position, displacement and direction algebraically, so that geometric problems can be solved by calculation. In H2 Mathematics, we extend vectors from two dimensions to three dimensions, and use them to work with lines and planes, angles, distances, intersections and the foot of the perpendicular. Outside mathematics, vectors are used for forces and velocities in physics, in navigation, in computer graphics and 3D modelling, and throughout engineering. Vectors also lie at the heart of mathematics in linear algebra, which has foundational applications in fields such as physics, engineering, computer science, data science, machine learning and artificial intelligence, economics and cryptography.



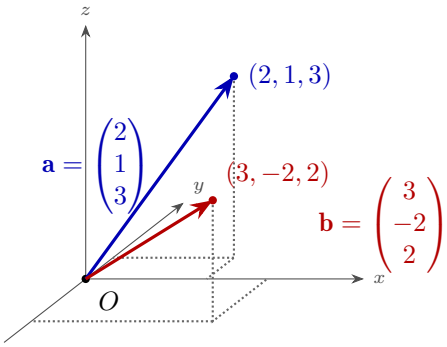
Name	Equation	Diagram
Negative of **a** $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$	
$2\mathbf{a}$	$k\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} kx \\ ky \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 4 \\ 2 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \end{pmatrix}\right\rvert = \sqrt{x^2 + y^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2 + 1^2}$ $= \sqrt{5}$	

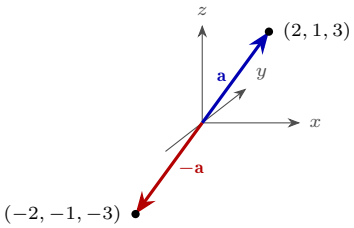
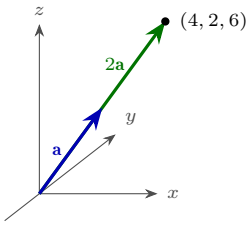
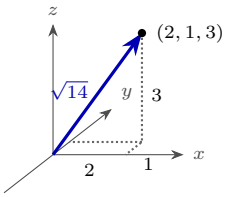
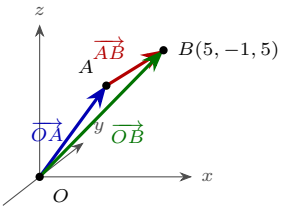
Name	Equation	Diagram
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = 3\begin{pmatrix} 2 \\ 1 \end{pmatrix} - 2\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $= \begin{pmatrix} 6 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 4-2 \\ 3-2 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ 1 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\,\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 3-1 \\ 3-2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 7-3 \\ 5-3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ $= 2\,\overrightarrow{AB}$	

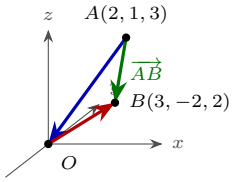
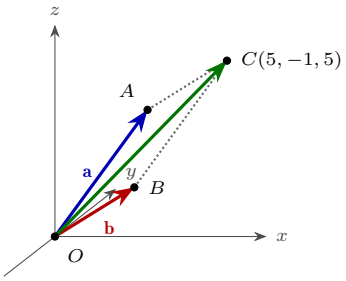
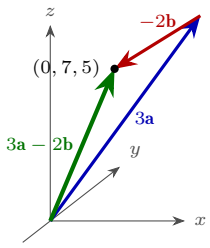
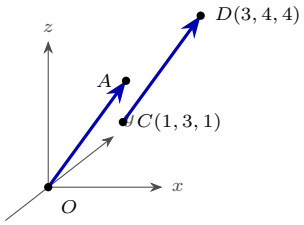
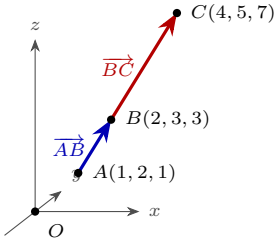
13.2 Vectors in Three Dimensions

In H2 Mathematics, we extend vectors to three dimensions.

$$\mathbf{a} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$



Name	Equation	Diagram
Negative of **a** $-\mathbf{a}$	$-\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -x \\ -y \\ -z \end{pmatrix}$ $-\mathbf{a} = \begin{pmatrix} -2 \\ -1 \\ -3 \end{pmatrix}$	
2a	$k\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} kx \\ ky \\ kz \end{pmatrix}$ $2\mathbf{a} = 2\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 6 \end{pmatrix}$	
Magnitude $\lvert\mathbf{a}\rvert$	$\left\lvert\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\rvert = \sqrt{x^2+y^2+z^2}$ $\lvert\mathbf{a}\rvert = \sqrt{2^2+1^2+3^2}$ $= \sqrt{14}$	
Triangle law	$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	

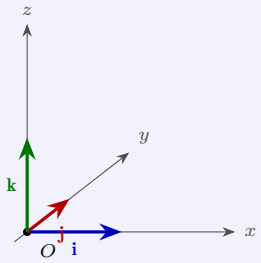
Name	Equation	Diagram
$\overrightarrow{AB}$	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}$	
Parallelogram law $\mathbf{a} + \mathbf{b}$	$\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ $\mathbf{a} + \mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 5 \\ -1 \\ 5 \end{pmatrix}$	
Linear combination	$\lambda\mathbf{a} + \mu\mathbf{b} = \begin{pmatrix} \lambda a_1 + \mu b_1 \\ \lambda a_2 + \mu b_2 \\ \lambda a_3 + \mu b_3 \end{pmatrix}$ $3\mathbf{a} - 2\mathbf{b} = \begin{pmatrix} 6 \\ 3 \\ 9 \end{pmatrix} - \begin{pmatrix} 6 \\ -4 \\ 4 \end{pmatrix}$ $= \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix}$	
Equal vectors	$\overrightarrow{OA} = \overrightarrow{CD}$ $\overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ $\overrightarrow{CD} = \begin{pmatrix} 3-1 \\ 4-3 \\ 4-1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$	
Collinear vectors	$\overrightarrow{AB} = k\,\overrightarrow{BC}$ $\Rightarrow A, B, C$ collinear $\overrightarrow{AB} = \begin{pmatrix} 2-1 \\ 3-2 \\ 3-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 4-2 \\ 5-3 \\ 7-3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$ $= 2\,\overrightarrow{AB}$	

13.3 Basic Terminology and Notation

Definition 13.3.1. Unit vectors i, j, k

i, **j** and **k** are the unit vectors in the positive directions of the x-, y- and z-axes:

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



Any vector can be written in terms of **i**, **j**, **k**:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Example 13.3.1

Write in column vector form: (a) $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ (b) $3\mathbf{i} + 2\mathbf{k}$. (c) Write $\begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix}$ in terms of $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Solution.

(a) $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}$ (b) $3\mathbf{i} + 2\mathbf{k} = 3\mathbf{i} + 0\mathbf{j} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$ (c) $\begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix} = 5\mathbf{i} - \mathbf{j}$

Theorem 13.3.1. Magnitude of a vector

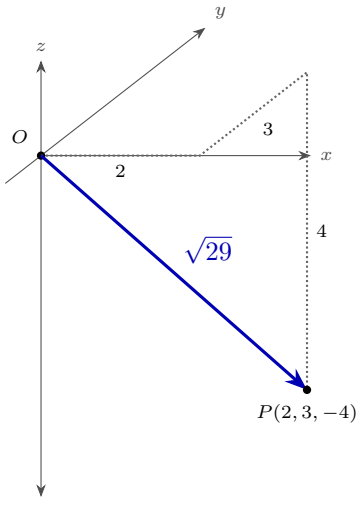
$$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad \Rightarrow \quad |\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$$

Example 13.3.2

Find $|2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}|$.

Solution.

$$\begin{aligned} |2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}| &= \sqrt{2^2 + 3^2 + (-4)^2} \\ &= \sqrt{4 + 9 + 16} \\ &= \sqrt{29} \end{aligned}$$



Theorem 13.3.2. Parallel vectors

$$\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} = \lambda\mathbf{b} \text{ for some scalar } \lambda \neq 0 \quad (\text{alternatively, } \mathbf{a} \times \mathbf{b} = \mathbf{0})$$

$\lambda > 0$: same direction; $\lambda < 0$: opposite directions.

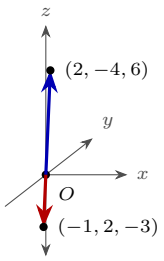
Example 13.3.3

(a) Show that $\begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$. (b) Find p and q such that $\begin{pmatrix} p \\ 6 \\ q \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$.

Solution.

(a) $$\begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} = -2 \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix} \parallel \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} \text{ (shown)}$$



(b) $$\begin{pmatrix} p \\ 6 \\ q \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

$$6 = 3\lambda \Rightarrow \lambda = 2$$

$$\therefore p = 2, \quad q = -4$$

Theorem 13.3.3. Unit vectors

The unit vector in the direction of $\mathbf{a}$ is

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Example 13.3.4

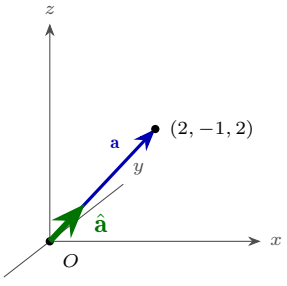
Let $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Find (a) the unit vector in the direction of $\mathbf{a}$, (b) the vectors of magnitude 6 parallel to $\mathbf{a}$.

Solution.

(a) $$|\mathbf{a}| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3$$

$$\hat{\mathbf{a}} = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

(b) $$\pm 6\hat{\mathbf{a}} = \pm \frac{6}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \pm \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix}$$



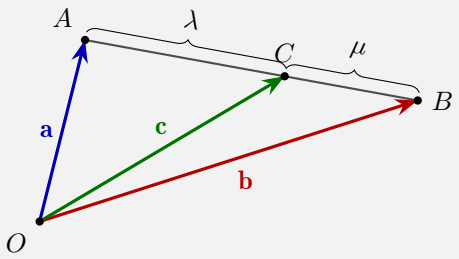
13.3.1 Ratio Theorem

Theorem 13.3.4. Ratio Theorem

If C lies on AB with $AC : CB = \lambda : \mu$, then

$$\overrightarrow{OC} = \frac{\mu\overrightarrow{OA} + \lambda\overrightarrow{OB}}{\lambda + \mu}$$

i.e. $\mathbf{c} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$.



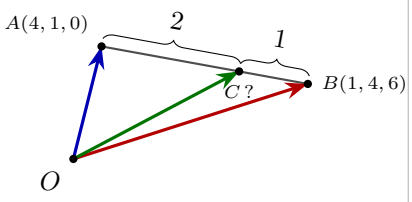
Example 13.3.5

$\overrightarrow{OA} = \begin{pmatrix}4\\1\\0\end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix}1\\4\\6\end{pmatrix}$. C lies on AB with $AC : CB = 2 : 1$. Find $\overrightarrow{OC}$.

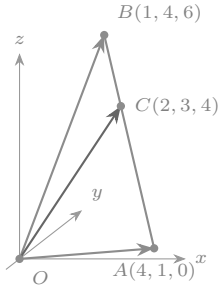
Solution.

$$\begin{aligned}\overrightarrow{OC} &= \frac{1\overrightarrow{OA} + 2\overrightarrow{OB}}{2+1}\\ &= \frac{1}{3}\left[\begin{pmatrix}4\\1\\0\end{pmatrix} + \begin{pmatrix}2\\8\\12\end{pmatrix}\right]\\ &= \frac{1}{3}\begin{pmatrix}6\\9\\12\end{pmatrix}\\ &= \begin{pmatrix}2\\3\\4\end{pmatrix}\end{aligned}$$

Diagram you can draw:



Actual diagram: (you do not need to draw this)



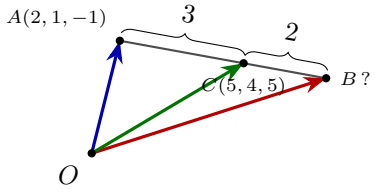
Example 13.3.6

A has position vector $\begin{pmatrix}2\\1\\-1\end{pmatrix}$ and C has position vector $\begin{pmatrix}5\\4\\5\end{pmatrix}$. C lies on AB with $AC : CB = 3 : 2$. Find $\overrightarrow{OB}$.

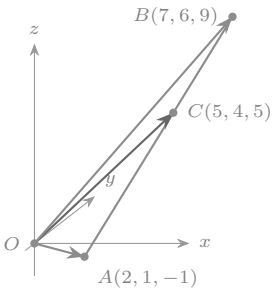
Solution.

$$\begin{aligned}\overrightarrow{OC} &= \frac{2\overrightarrow{OA} + 3\overrightarrow{OB}}{3+2}\\ \therefore\ \overrightarrow{OB} &= \frac{5\overrightarrow{OC} - 2\overrightarrow{OA}}{3}\\ &= \frac{1}{3}\left[\begin{pmatrix}25\\20\\25\end{pmatrix} - \begin{pmatrix}4\\2\\-2\end{pmatrix}\right]\\ &= \frac{1}{3}\begin{pmatrix}21\\18\\27\end{pmatrix}\\ &= \begin{pmatrix}7\\6\\9\end{pmatrix}\end{aligned}$$

Diagram you can draw:



Actual diagram: (you do not need to draw this)



13.4 Scalar Product

In this section, we learn about how we can multiply vectors together. There are two different ways we can do so, namely the Scalar Product and Vector Product. The Scalar and Vector products each have their own respective applications in the study of Vectors. We will learn about these applications more in depth later in this section.

13.4.1 Scalar (Dot) Product

Definition 13.4.1. Scalar (dot) product

The scalar or dot product of two vectors $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ is defined as

$$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_1b_1 + a_2b_2 + a_3b_3.$$

Example 13.4.1

Find $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix}$.

Solution.

$$\begin{aligned} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix} &= 1(5) + 2(-1) + 3(3) \\ &= 12 \end{aligned}$$

? **Thinking Question:** Note that the Scalar Product yields a result that is scalar (a magnitude). What does this suggest about the other form of multiplication (Vector Product)?

Problem 13.4.2

It is given that $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -3 \\ 2 \\ 7 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 4 \\ -5 \\ -1 \end{pmatrix}$. Find the following

a) $\mathbf{a} \cdot \mathbf{b}$ c) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c})$ e) $\mathbf{a} \cdot (2\mathbf{c})$ g) $\mathbf{a} \cdot \mathbf{a}$

b) $\mathbf{b} \cdot \mathbf{a}$ d) $\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ f) $(2\mathbf{a}) \cdot \mathbf{c}$

Ans: a) 22 b) 22 c) 13 d) 13 e) -18 f) -18 g) 14

Learning Points:

- Parts (a) and (b) show that the order of the vectors in the scalar product does NOT matter. In other words, the operation is commutative, [i.e. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$]
- Parts (c) and (d) show that the scalar product operation is distributive over vector addition, [i.e. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$]
- Parts (e) and (f) show that the scalar multiplication of vectors is associative over the scalar product, [i.e. $(k\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (k\mathbf{b})$]

13.4.2 Basic Properties of Scalar Product

Taking reference to the problems in the previous section (Scalar Product), we can see that the Scalar Product has properties that are similar to basic algebra multiplication. In general, we have the following properties of the Scalar Product.

Theorem 13.4.1. Laws of scalar product

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$$
(Commutative Law of Scalar Product)

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}, \quad (\mathbf{b} + \mathbf{c}) \cdot \mathbf{a} = \mathbf{b} \cdot \mathbf{a} + \mathbf{c} \cdot \mathbf{a}$$
(Distributive Law of Scalar Product)

$$\mathbf{a} \cdot (\lambda \mathbf{b}) = \lambda(\mathbf{a} \cdot \mathbf{b}) = (\lambda \mathbf{a}) \cdot \mathbf{b} \quad \lambda \in \mathbb{R}$$
(Associative Law of Scalar Product)

$$|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a}$$

Notes:

1. The scalar product $\mathbf{a} \cdot \mathbf{b}$ is indeed a scalar.
2. $(\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ is a scalar multiple of the vector $\mathbf{c}$. The scalar is $\mathbf{a} \cdot \mathbf{b}$.
3. It is meaningless to write $(\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{c}$ (Why?)
4. If $\mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$, then $\mathbf{a} \cdot \mathbf{b} = 0$.

?

Questions:

- Does the cancellation law hold: $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} \implies \mathbf{b} = \mathbf{c}$?
- Is the following true: $\mathbf{a} \cdot \mathbf{b} = \mathbf{c} \cdot \mathbf{d} \implies \mathbf{a} = \dfrac{\mathbf{c} \cdot \mathbf{d}}{\mathbf{b}}$?
- Is the following true: $\mathbf{a} \cdot \mathbf{b} = 0 \implies \mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$?

Justify your answer.

Note: All of the above are not true.

13.4.3 Angle Between Two Vectors

We now look at how we are able to use the Scalar Product to find the angle between two vectors. This is one of the applications of the Scalar Product. To be clear, when we say the angle θ between two vectors, we are referring to the non-reflex angle between them, where either both their start points coincide, or end points coincide.

Correct angle:



Wrong angle:



In order to find this angle, we make use of the following formula.

Definition 13.4.2. (Definition of Scalar Product)

The scalar product of two vectors $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta,$$

where θ is the angle between two vectors $\mathbf{a}$ and $\mathbf{b}$.

Example 13.4.3

Find the angle between the following vectors $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$.

Solution.

$$\begin{aligned}
\mathbf{a} \cdot \mathbf{b} &= |\mathbf{a}||\mathbf{b}| \cos\theta \\
\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} &= \left| \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right| \left| \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} \right| \cos\theta \\
-2 + 4 + 3 &= \sqrt{14}\sqrt{9} \cos\theta \\
\frac{5}{3\sqrt{14}} &= \cos\theta \\
\theta &= \cos^{-1}\left(\frac{5}{3\sqrt{14}}\right) \\
\theta &= 63.5^\circ
\end{aligned}$$

Hence we can use Definition 13.4.2 to find the non-reflex angle θ between vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}.$$

Problem 13.4.4

1) Points O, A, B are such that $\overrightarrow{OA} = \mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$ and $\overrightarrow{OB} = 5\mathbf{i} - \mathbf{j}$

   (a) Find the size of angle AOB

   (b) Find the size of angle OAB.

   Ans: (a) 87.8° (b) 133.9°

2) It is given that $\mathbf{c} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$, $\mathbf{d} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and $\mathbf{e} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and

   (a) the vector $\mathbf{d}$

   (b) the vector $\mathbf{e}$

   (c) the positive y-axis

   Ans: (a) 57.7° (b) 90° (c) 64.6°

3) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:

$$|\mathbf{a}||\mathbf{b}| = 6.$$

   Find the value of $\mathbf{a} \cdot \mathbf{b}$ for each of the following scenarios.

   (a) The angle between $\mathbf{a}$ and $\mathbf{b}$ is 60°.

   (b) The angle between $\mathbf{a}$ and $\mathbf{b}$ is 120°.

   (c) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

   Ans: (a) $\mathbf{a} \cdot \mathbf{b} = 3$ (b) $\mathbf{a} \cdot \mathbf{b} = -3$ (c) $\mathbf{a} \cdot \mathbf{b} = 0$

Sign of the Scalar Product

From Problem 3, what do you observe about the sign of the scalar product $\mathbf{a} \cdot \mathbf{b}$ when the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute or obtuse? Why is this so?

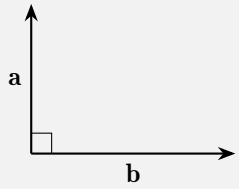
Perpendicular Vectors

From 3(c), we see that $\mathbf{a} \cdot \mathbf{b} = 0$. Is this always the case when $\mathbf{a}$ and $\mathbf{b}$ are perpendicular? Why? We will explore more on this in the next section.

13.4.4 Perpendicular Vectors

We can see that from the previous results, that the scalar product of two non-zero vectors that are perpendicular to each other is zero.

Theorem 13.4.2. Perpendicular vectors



If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular vectors, then

$$\mathbf{a} \cdot \mathbf{b} = 0$$

Note: This result can be used to test for two perpendicular vectors, provided that the two vectors are non-zero.

Example 13.4.5

Show that the vector $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ are perpendicular to each other.

Solution.

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} \\ &= -1 + 2 - 1 \\ &= 0 \end{aligned}$$

Since $\mathbf{a} \cdot \mathbf{b} = 0$, $\mathbf{a}$ & $\mathbf{b}$ are perpendicular.

Problem 13.4.6

1) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$ respectively. Show that OA is perpendicular to OB.

2) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a} = 2p\mathbf{i} + 3p\mathbf{j} + 6p\mathbf{k} \quad \text{and} \quad \mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$$

where $p > 0$. It is given that $|\mathbf{a}| = |\mathbf{b}|$.

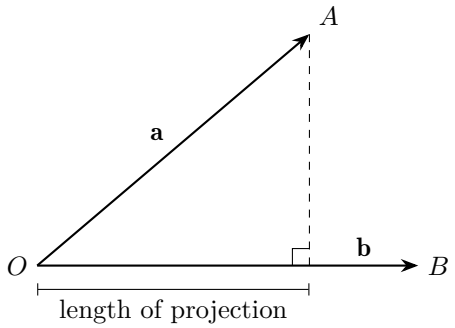
(i) Find the exact value of p.

(ii) Show that the vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are perpendicular.

Ans: (i) $p = \dfrac{3}{7}$

13.4.5 Length of Projection & Projection Vector

The projection vector of a vector **a** onto **b** is the vector component of **a** that is parallel to **b**. The magnitude (or length) of this vector is known as the length of projection. To help you understand this better, imagine you are shining a torchlight from a point directly above **a**. The length of projection is the length of the shadow that is cast on **b**.



In order to find the length of projection and projection vector of **a** onto **b**, we can make use of the following result.

Theorem 13.4.3. Projection of Vectors

For any two non-zero vectors **a** and **b**,

$$\text{length of projection of } \mathbf{a} \text{ onto } \mathbf{b} = |\mathbf{a} \cdot \hat{\mathbf{b}}|,$$
$$\text{projection vector of } \mathbf{a} \text{ onto } \mathbf{b} = (\mathbf{a} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}}.$$

? **Thinking Question:** *What is the difference between considering the projection of* **a** *onto* **b** *and* **b** *onto* **a***?*

Example 13.4.7

The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}.$$

Find the length of projection and the projection vector of the vector $\mathbf{a}$ onto $\mathbf{b}$.

Solution.

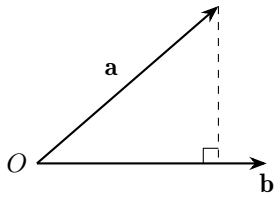
$$\text{Length of projection} = |\mathbf{a} \cdot \hat{\mathbf{b}}|$$

$$\hat{\mathbf{b}} = \frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$$

$$\therefore \text{ length} = \left| \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \right|$$

$$= \frac{1}{3}\left| \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix} \right|$$

$$= \frac{1}{3} \text{ units}$$



$\therefore$ The length of projection of the vector $\mathbf{a}$ onto $\mathbf{b}$ is $\frac{1}{3}$ units. The projection vector is

$$\frac{1}{3}\left(\frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}\right) = \frac{1}{9}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$$

Problem 13.4.8

1) Given that $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 2\mathbf{k} - \mathbf{k}$, find the length of projection and the projection vector of

   (a) $\mathbf{a}$ onto $\mathbf{b}$,

   (b) $\mathbf{b}$ onto $\mathbf{a}$.

2) With reference to the origin O, the points A, B and C have position vectors given by

$$\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}, \ \mathbf{b} = 3\mathbf{i} + \mathbf{k} \text{ and } \mathbf{c} = 2\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$$

   respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

Ans:

1a) $\frac{2}{\sqrt{6}}, \frac{1}{3}\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ 1b) $\frac{2}{3}, \frac{2}{9}\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$

2) $\frac{2}{3}$